

SGA 2: Local Cohomology and Lefschetz Theorems

Exposé VIII: The Finiteness Theorem

Bounded source-aligned working unit — 19 July 2026

Authority. Corrected French lines 2661–2668; original printed p. 89; physical source-PDF p. 79; recomposed running p. 71. The compiled reader is the output of the same corrected French edition, not an independent original typescript witness. Line 2669 is blank; Corollary 2.3 begins at line 2670 and is excluded.

Corollary 2.2⁽³⁾. *Under the hypotheses of the preceding theorem, condition a) is equivalent to:*

c) *for every $x \in U$ such that $c(x) = 1$, one has $H^{i-1}(\mathcal{F}_x) = 0$.*

⁽³⁾*Editor's note.* Strictly speaking, this is a corollary of the proof that follows, not of the statement. The implication c) $\Rightarrow$ a) is tautological. The converse is not, but follows from the proof. More precisely, as below, one covers X by open subsets embeddable in regular schemes; as explained below, this permits one to reduce to the case where $X = \text{Spec}(A)$ is affine and regular and $\mathcal{F} = \widetilde{M}$, where M is an A -module of finite projective dimension. In this case it is shown that conditions a) and c) are equivalent to the dual conditions a') and c'). One then shows that c') implies condition d) (see below), which in turn implies a'). See the discussion following 2.4.